

VEZA

LEARNING CLOUD

MASTER IMPLEMENTATION PLAN

Product delivery, engineering, migration, quality, operations and institutional rollout

Prepared for Michael Ndhlovu
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Version 1.0

Document control

Field	Value
Document	Veza Learning Cloud Master Implementation Plan
Source architecture	Veza Learning Cloud Product, Experience, Data and Technical Architecture v1.0
Purpose	Translate the approved target architecture into a buildable, testable and commercially deployable programme of work.
Planning horizon	Approximately 30 months to general availability, subject to team size, procurement, pilot availability and scope control.
Planning method	Outcome-based phases, dependency gates, vertical slices, architecture decision records, risk-led testing and institution co-design.
Primary decision authority	Veza Product and Architecture Council
Change control	Material scope, architecture, compliance, security, timeline or commercial changes require a recorded decision and impact assessment.

How to use this plan

This plan is the execution companion to the architecture document. It is not a fixed feature checklist. It defines the order in which capabilities must be proven, the cross-cutting controls that apply to every module, and the evidence required before work can progress to the next release gate.

- Product leaders use the phase outcomes, scope boundaries and acceptance gates to control roadmap decisions.
- Engineering teams break the epics into sprint-sized vertical slices while preserving the stated dependencies and invariants.
- Design and research teams use the journey validation and UX acceptance criteria to test usability with each institution segment.
- Security, privacy and quality teams use the control matrices as release-blocking criteria, not post-launch remediation lists.
- Implementation and customer-success teams use the migration, readiness and rollout sections to onboard institutions safely.

Planning assumptions

Assumption	Planning implication
Delivery model	A dedicated cross-functional product organisation is available. A small agency-style team cannot safely deliver the complete scope in one release.
Architecture	Begin as a modular monolith with strict bounded-context ownership, transactional outbox and service-ready interfaces. Split services only when scaling, isolation or team ownership justifies it.
Cloud	AWS is the reference deployment environment. Infrastructure remains reproducible and region-aware through infrastructure as code.
Clients	Responsive web is the primary experience. Native mobile begins after the web domain and offline contracts have stabilised.
Market	The first reference tenants should represent at least two operating models, such as a training academy and a tertiary or school-like institution.
Standards	Interoperability is built from canonical domain models and adapters. Veza does not make external standards its internal source of truth.
AI	AI assistance is optional, governed and never authoritative for grades, admissions, disciplinary outcomes or safeguarding decisions.
Timeline	The duration assumes parallel workstreams and staged pilots. It is not a commitment until team capacity and tenant commitments are baselined.

Executive implementation strategy

Veza should be delivered as a sequence of institutional operating capabilities, not as isolated screens. Each phase must produce an internally coherent vertical slice that can be observed, audited, supported and tested with real institutional workflows. The programme begins with tenant and identity trust boundaries, then establishes the academic graph, then adds day-to-day teaching, institutional operations and high-stakes assessment. Interoperability, intelligence and commercial scale follow only after the canonical workflows are stable.

Principle	Execution rule
Trust before breadth	No learning feature ships before tenant isolation, authorisation, audit, retention classification and operational telemetry exist for it.
Vertical slices	A slice includes UI, API, domain rules, events, permissions, audit, analytics, tests, documentation and support diagnostics.
Institutional configurability	Configurable terminology, policy and workflow states are preferred over customer-specific forks.
Human accountability	Automation may recommend, validate or queue work, but accountable users approve sensitive academic decisions.
Version everything consequential	Course blueprints, rubrics, grade formulas, policy rules, credentials, consent and published results require effective-dated versions.
Evidence-driven gates	A phase is complete only when its acceptance evidence exists. Percentage-complete reporting is secondary.
Operational sameness	Shared and dedicated tenants use the same control plane, release train, observability model and support processes.
Low-bandwidth resilience	Critical learner and teacher flows must tolerate constrained devices, intermittent connectivity and asynchronous work.

Programme outcome and release train

Release line	Audience	Cadence	Purpose
Internal edge	Veza staff and synthetic tenants	Continuous	Early integration, destructive testing and platform operations validation.
Design partner	Named institutions under pilot agreement	Every 2-4 weeks	Workflow validation, migration rehearsals and controlled operational use.
Preview	Approved early adopters	Every 4-6 weeks	Broader compatibility and scale testing with contractual support boundaries.
General availability	Production customers	Monthly feature train; urgent fixes as required	Stable, documented and supported releases with tenant-ring controls.
Long-term support	Dedicated or regulated tenants	Quarterly roll-up where contracted	Predictable upgrade windows while preserving the unified product model.

Master milestones

ID	Milestone	Exit evidence
M0	Architecture mobilisation complete	Product charter, delivery model, ADR process, environments, design foundations and security baseline approved.
M1	SaaS foundation operational	Tenant provisioning, identity, scoped authorisation, entitlements, audit, observability and deployment pipeline pass isolation tests.
M2	Academic core usable	Institution structure, periods, catalogue, course blueprints, runs, cohorts, enrolments and calendars support a complete setup flow.
M3	Teaching and learning pilot-ready	Studio, learner course room, assignments, submissions, gradebook, announcements, attendance and progress operate end to end.
M4	Institution operations pilot-ready	Admissions, roster operations, timetabling, guardian/sponsor access, intervention cases, reports and SSO are production-capable.
M5	Assessment governance complete	Item bank, QTI, exams, moderation, ratification, appeals, accommodations and controlled grade correction pass high-stakes testing.
M6	Ecosystem and portability complete	LTI, OneRoster, Common Cartridge, APIs, webhooks and credentials pass conformance-oriented contract tests.
M7	Commercial scale readiness	Metering, plans, marketplace controls, data warehouse, AI governance, dedicated tier automation and customer operations are ready.
M8	General availability	Reference tenants migrated, support and DR rehearsals passed, SLOs proven and launch approval signed.

1. Programme governance and operating model

1.1 Decision forums

Forum	Cadence	Decision scope	Required participants
Product and Architecture Council	Weekly	Roadmap, domain boundaries, cross-cutting policies, major ADRs and phase-gate recommendations.	Product lead, chief architect, design lead, engineering leads, security/privacy and delivery lead.
Academic Advisory Group	Fortnightly during discovery; monthly thereafter	Academic validity, assessment governance, safeguarding, accessibility and institution-segment differences.	Academic leaders, registrar/exam representatives, instructors, accessibility specialist and product.
Security and Privacy Review	Fortnightly	Threat model changes, DPIAs, data classification, incident readiness, vendor risk and privileged access.	Security, privacy/legal, platform engineering and affected product lead.
Delivery Review	Weekly	Milestones, dependencies, blockers, quality trends, staffing and scope trade-offs.	Delivery lead, workstream leads, product operations and QA.
Release Readiness Board	Per release	Go/no-go based on test, security, migration, support and rollback evidence.	Release manager, engineering, SRE, QA, security, support and product owner.
Tenant Change Advisory	As required	Tenant-visible breaking changes, migration windows and dedicated-tier exceptions.	Customer success, implementation, product, SRE and affected tenant representatives.

1.2 Core roles and accountability

Role	Accountability
Executive sponsor	Funding, commercial direction, institutional partnerships and escalation authority.
Product director	Product outcomes, roadmap, market fit, phase scope and prioritisation.
Chief architect	Domain boundaries, integration patterns, non-functional requirements and technical coherence.
Design director	Research, information architecture, design system, accessibility and product experience quality.
Engineering managers	Team health, delivery capability, technical execution and engineering standards.

Role	Accountability
Domain product managers	Backlog and acceptance for assigned bounded contexts.
Security and privacy lead	Threat models, security controls, privacy-by-design, vendor assurance and incident readiness.
Academic product specialist	Academic terminology, assessment policy, regulatory variance and workflow validation.
SRE/platform lead	Cloud platform, CI/CD, observability, SLOs, capacity, DR and operational readiness.
Quality engineering lead	Test strategy, automation, release evidence, performance, accessibility and workflow assurance.
Implementation lead	Tenant discovery, configuration, migration, training and go-live readiness.
Customer success lead	Adoption, health, support escalation and renewal outcomes.
Data/analytics lead	Canonical event model, warehouse, metrics, data quality and analytics governance.

1.3 Architecture decision records

Every decision that changes a domain boundary, data ownership, tenancy model, security property, integration contract, deployment topology or irreversible vendor dependency must be captured as an ADR. Each ADR records context, considered options, decision, consequences, migration path, owner and review date. ADRs are versioned with the codebase and linked to affected epics and runbooks.

1.4 Delivery artefacts required for every epic

- Problem statement and measurable user outcome.
- Journey map, edge cases and permission matrix.
- Domain model changes and invariants.
- API/event contracts and idempotency behaviour.
- Data classification, retention and audit requirements.
- Threat-model delta and abuse cases.
- Accessibility and low-bandwidth acceptance criteria.
- Observability signals, support diagnostics and runbook updates.
- Automated test plan, migration plan and rollback plan.
- Customer-facing documentation and operational ownership.

2. Delivery organisation and workstreams

2.1 Recommended team topology

Team	Primary ownership	Indicative composition at scale
Platform and Control Plane	Tenant lifecycle, entitlements, billing hooks, identity integration, authorisation platform, audit and admin operations.	1 PM, 1 design partner, 1 EM/lead, 4-6 engineers, shared QA/SRE/security.
Academic Core	Institutions, academic structure, periods, catalogue, curricula, offerings, runs, cohorts, enrolments and progression.	1 PM, 1 product designer, 1 lead, 5-7 engineers, 1-2 QEs.
Studio and Learning Experience	Authoring, templates, content blocks, course room, learner experience, communications and resource delivery.	1 PM, 2 designers/researchers, 1 lead, 6-8 engineers, 2 QEs.
Assessment and Outcomes	Assignments, submissions, item bank, exams, rubrics, gradebook, moderation, appeals, credentials.	1 PM, academic specialist, 1 designer, 1 lead, 6-8 engineers, 2 QEs.
Institution Operations	Admissions, timetabling, attendance, guardian/sponsor, interventions, cases and institutional reporting.	1 PM, 1 designer, 1 lead, 5-7 engineers, 1-2 QEs.
Ecosystem and Data	Integrations, standards adapters, APIs/webhooks, search, analytics pipeline, warehouse and reporting.	1 PM, 1 lead, 5-7 engineers/data engineers, 1 QE.
Platform Engineering and SRE	Infrastructure, environments, CI/CD, observability, media, resilience, DR, cost and developer platform.	1 platform lead, 4-6 platform/SRE engineers, security partner.
Implementation and Customer Operations	Tenant discovery, migration, configuration, training, support, knowledge base and health monitoring.	Implementation lead, 2-6 consultants, support lead, support engineers, training/content specialist.

2.2 Minimum viable delivery organisation

For the first 12 months, Veza can operate with four product squads rather than eight: Platform, Academic Core, Learning Experience, and Assessment/Operations, supported by a small platform/SRE group and shared design, security, QA and implementation specialists. Below this level, the programme must narrow scope rather than compress quality activities.

Capability	Recommended minimum
Product and delivery	1 product director, 3 domain product managers, 1 delivery/programme lead, 1 academic product specialist
Design and research	1 design director, 3 product designers, 1 UX researcher/content designer; accessibility expertise retained

Capability	Recommended minimum
Engineering	1 chief architect, 4 technical leads, 16-22 software engineers, 3 platform/SRE engineers, 2 data/integration engineers
Quality	1 QE lead, 4-6 quality engineers with automation, accessibility and performance coverage
Security/privacy	1 security lead plus privacy/legal support; specialist penetration testing contracted
Implementation/support	1 implementation lead, 2 implementation consultants, 1 support lead and 2 support engineers by first production pilot

2.3 Workstream dependency rules

Dependency	Rule
Identity before enrolment	Users and memberships must have stable global and tenant identities before admissions or enrolment creates academic records.
Academic graph before learning UI	Programmes, offerings, versions, runs, cohorts and enrolments are canonical. Course-room screens cannot invent parallel structures.
Content versioning before approvals	Studio versioning and immutable approval snapshots precede formal course approval and run upgrades.
Assessment model before gradebook	Attempts, rubrics, evidence, accommodations and marking states are established before aggregate grade calculations.
Audit before privileged operations	No grade override, assisted session, export, role assignment or data merge ships without immutable audit.
Events before analytics	Canonical domain events are specified alongside commands; dashboards do not scrape transactional tables ad hoc.
Supportability before pilots	Tenant diagnostics, feature flags, incident correlation IDs, runbooks and rollback must exist before design-partner use.
Migration before go-live	Every production tenant completes at least one dry-run import, reconciliation and rollback rehearsal.

3. Integrated roadmap and sequencing

The roadmap uses nine implementation stages. Stages overlap only where their input contracts are stable. The month ranges are planning bands, not promises; gate evidence determines progression.

Stage	Indicative months	Primary outcome	Pilot status
0. Mobilise and de-risk	0-2	Decisions, team, discovery, foundations and risk baselines.	No tenant operations
1. SaaS foundation	2-6	Secure tenant control plane and developer platform.	Synthetic tenants only
2. Academic core	4-9	Canonical institutional and academic graph.	Configuration pilot
3. Core learning	7-14	Create, enrol, teach, submit, mark and track.	Design-partner pilot
4. Institutional operations	11-18	Run admissions, schedules, attendance, support and reporting.	Operational pilot
5. Assessment rigour	14-21	High-stakes assessment and governance.	Controlled assessment pilot
6. Ecosystem and credentials	18-24	Portability, standards and external tools.	Integration pilot
7. Intelligence and commercial scale	21-28	Analytics, AI, metering, marketplace and dedicated tiers.	Preview customers
8. GA hardening and rollout	26-30	Reliability proof, migration factory and general availability.	GA launch

3.1 Critical path

1. Tenant context, identity, scoped authorisation, audit and entitlements.
2. Institution structure, terminology, academic periods and people relationships.
3. Catalogue, curriculum, blueprint versioning, course runs, cohorts and enrolments.
4. Studio publishing, learner course room, assignments, submissions and gradebook.
5. Attendance, timetable, progress and intervention workflows.
6. Assessment governance, moderation, ratification, appeals and credentials.
7. Interoperability adapters, migration tooling and reference-tenant rollout.
8. Operational scale, commercial metering, analytics and dedicated deployment automation.

3.2 Scope control

Each phase has an explicit “not yet” boundary. Requests outside the boundary enter the product discovery backlog unless they are necessary to satisfy security, legal, data integrity or pilot viability. Customer-specific forks are prohibited. Where a reference tenant requires variance, the team must determine whether it is configurable policy, a supported extension point, an integration, or genuinely out of product scope.

Stage 0 - Mobilise and de-risk

Planning band: Months 0-2. Outcome: Create the execution system, validate the highest-risk assumptions, select design partners and establish measurable product and engineering baselines.

Delivery scope

Work package	Detailed implementation
Product and market validation	Interview 12-20 institutions across target segments; map current systems, migration constraints, purchasing roles, academic calendars, regulatory obligations and failure pain. Select two design partners with different operating models.
Domain validation	Run event-storming workshops for admissions-to-credential, course-authoring-to-teaching, assessment-to-appeal and tenant-provisioning-to-offboarding. Confirm canonical vocabulary and tenant relabelling rules.
Experience foundations	Test navigation models for learner, instructor, academic operations and tenant administrator. Produce responsive shell prototypes, workspace context switching and bento dashboard information hierarchy.
Architecture foundations	Approve modular-monolith boundary map, repository strategy, environment model, tenancy model, identity model, event/outbox pattern, storage, search and media abstractions.
Security/privacy baseline	Complete threat model, data classification catalogue, preliminary DPIA, subprocessor strategy, privileged-access model and incident classification.
Delivery system	Create backlog hierarchy, ADR template, coding standards, branch/release model, definition of ready/done, test pyramid and release evidence template.

Exit gate

- Signed product charter and target segments
- Approved bounded-context map and top 20 ADRs
- Design-partner agreements and research schedule
- Threat model and privacy baseline
- Working monorepo, CI checks and ephemeral environment proof

Explicitly deferred unless required for the gate

- Production customer data
- Native mobile development
- Marketplace or generative AI
- Customer-specific custom code

Stage 1 - SaaS foundation

Planning band: Months 2-6. Outcome: Deliver the trustworthy platform substrate on which every institutional capability depends.

Delivery scope

Work package	Detailed implementation
Control plane	Tenant creation, states, plans, module entitlements, quotas, regions, branding pointers, custom-domain lifecycle and tenant offboarding workflow.
Identity	Global user identity, tenant memberships, invitation, passwordless/password options, MFA, session management, recovery, SSO configuration skeleton and identity-linking rules.
Authorisation	Permission catalogue, roles, scoped assignments, policy conditions, delegated administration, field-level restrictions, explicit deny and policy decision logging.
Audit and compliance	Append-only audit events, actor and tenant context, before/after references, export receipts, assisted-session controls, retention labels and legal holds.
Platform engineering	AWS accounts/environments, IaC, networking, WAF, secrets, KMS, RDS/PostgreSQL, Redis, S3, CDN, queues/events, OpenSearch baseline, backup and observability.
Developer platform	Monorepo tooling, schema migration discipline, API standards, contract testing, feature flags, local development, seed tenants and synthetic data.
Design system	Tokens, typography, colour, density, grids, navigation shell, forms, data tables, command palette, empty/loading/error states and accessible component primitives.
Operations	Tenant-aware logs and traces, health dashboards, runbooks, incident tooling, support diagnostics, rate limits and cost allocation tags.

Exit gate

- Automated tenant provisioning and suspension
- Cross-tenant isolation property tests pass
- Privileged workflows produce immutable audit
- Restore from backup proven in non-production
- Design system passes keyboard and screen-reader baseline
- Release canary and rollback demonstrated

Explicitly deferred unless required for the gate

- Full academic workflows
- External standards certification

- Dedicated infrastructure variants beyond a proof of provisioning

Stage 2 - Academic core

Planning band: Months 4-9. Outcome: Establish one canonical graph for institutions, people and academic delivery.

Delivery scope

Work package	Detailed implementation
Institution structure	Institutions, campuses, faculties/departments, organisational units, terminology packs and effective-dated hierarchy.
People and relationships	Tenant profiles, identifiers, contacts, learner/staff status, guardian/sponsor/advisor relationships, emergency and consent boundaries.
Academic calendar	Academic years, terms, periods, holidays, blackout windows, time zones and tenant calendar overlays.
Catalogue and curriculum	Programmes, subjects/modules, credits/hours, outcomes/competencies, prerequisites, equivalencies and progression rules.
Blueprints and runs	Course blueprint versions, approval states, run creation, overlays, cohorts, groups, sections, staff allocation and capacities.
Enrolment	Applications-to-enrolment handoff contract, direct/bulk enrolment, waitlist, status history, effective dating, transfers, withdrawals and reinstatement.
Calendar foundation	Canonical event model, recurrence, participant/resource links, visibility and conflict validation.
Search	Tenant-scoped indexing for people, courses, runs and resources with permission-aware filtering.

Exit gate

- Complete institution setup without database intervention
- Create approved blueprint and run from UI/API
- Move learner between classes without losing history
- Detect capacity, prerequisite and schedule conflicts
- Terminology changes do not alter canonical APIs
- Academic entity history is effective-dated and auditable

Explicitly deferred unless required for the gate

- High-stakes exam delivery
- Advanced analytics
- Marketplace
- Native mobile

Stage 3 - Core learning

Planning band: Months 7-14. Outcome: Support the complete daily loop: create learning, deliver it, submit evidence, mark it and show progress.

Delivery scope

Work package	Detailed implementation
Veza Studio	Block editor, templates, modules/lessons, outcomes mapping, resources, autosave, collaborative comments, version diff, accessibility checks and publishing.
Course room	Learner home, module sequence, completion rules, resources, activities, announcements, discussions, bookmarks and accessible media playback.
Assignments	Task authoring, rubric, outcomes, attempts, dates, late rules, groups, accommodations, originality provider abstraction and notifications.
Submission service	Upload intents, quarantine/malware scan, immutable attempts, receipts, resumability, text/media/file submissions and offline-safe status.
Marking	Marker allocation, rubric grading, anchored feedback, drafts, release gates and controlled post-release correction.
Gradebook basics	Grade items, categories, formulas, exclusions, weighting, preview, learner views and export, all versioned.
Teaching workspace	Today dashboard, class context, lesson plan, session launch, poll/activity hooks, follow-up notes and learner concerns.
Communications	Announcements, targeted messages, templates, delivery preferences, digest, read state and communication audit.
Progress	Activity completion, outcome evidence, course progress, freshness indicators and instructor review queues.
Media	Upload, transcode, captions/transcripts, signed playback, thumbnails, bandwidth variants and provider abstraction.

Exit gate

- Author-to-published-course workflow passes review and rollback
- Learner completes a course on mobile and constrained network
- Submission reconnect does not duplicate attempts
- Marker and moderator roles cannot exceed scope
- Grade formula changes preview impact before publish
- Accessibility audit blocks critical authoring defects

Explicitly deferred unless required for the gate

- Automated admissions
- Complex exam proctoring
- External credential wallets
- Predictive risk scoring

Stage 4 - Institutional operations

Planning band: Months 11-18. Outcome: Enable institutions to operate repeatable academic administration beyond course delivery.

Delivery scope

Work package	Detailed implementation
Admissions	Application forms, evidence, duplicate matching, workflow states, decisions, waitlists, offers, consent and enrolment conversion.
Bulk data operations	CSV templates, mapping, validation, dry run, idempotent import, exception handling, reconciliation and rollback.
Timetabling	Session types, rooms/resources, staff availability, recurrence, clashes, substitutions, publication and change notifications.
Attendance	Manual, code/QR and integration-assisted evidence, late/excused rules, corrections, approval and attendance analytics.
Guardian/sponsor/employer	Scoped linked-person access, delegated consent, communications, progress summaries and relationship expiry.
Interventions and cases	Risk signal review, case creation, owners, actions, due dates, sensitive notes, escalation, outcomes and safeguarding separation.
Institution reports	Operational dashboards, scheduled reports, exports, data dictionary, report permissions and asynchronous generation.
Enterprise identity	SAML/OIDC SSO, SCIM provisioning, group mapping, domain discovery, break-glass access and certificate rotation.
Service operations	Tenant configuration diff, health score, support case links, diagnostic bundle and operational readiness dashboard.

Exit gate

- Admission to active learner completes without duplicate identity
- Published timetable changes reconcile all participants
- Attendance correction preserves original and reason
- Guardian cannot view unrelated or restricted information
- High-risk cases have explicit ownership and escalation
- SSO outage has documented break-glass and recovery

Explicitly deferred unless required for the gate

- Full finance/ERP replacement
- AI-driven admissions decisions
- Biometric attendance as official evidence without review

Stage 5 - Assessment rigour

Planning band: Months 14-21. Outcome: Deliver credible high-stakes assessment with separation of duties, evidence and appeals.

Delivery scope

Work package	Detailed implementation
Item bank	Versioned questions, metadata, outcomes, difficulty, exposure, ownership, review, lifecycle, pools and reuse.
QTI adapter	Import/export compatibility report, extension preservation, rendering tests and unsupported-feature handling.
Assessment assembly	Blueprint constraints, randomisation, equivalence, preview, forms, security classification and publication.
Exam delivery	Candidate roster, identity method, device/system check, time zones, autosave, reconnect, incident capture, accommodations and receipts.
Marking models	Anonymous, double, panel and sampled marking; allocation balancing; reconciliation and confidence flags.
Moderation	Policy-driven sampling, discrepancy thresholds, second review, approval chain and evidence pack.
Results governance	Calculation version, ratification, release windows, embargo, correction, learner notification and transcript impact.
Appeals	Submission, grounds, evidence, reviewer independence, decision, remediation and immutable outcome.
Accommodations	Effective dates, confidential source record, delivery instructions, extra time, breaks, formats and audit.
Assessment security	Question exposure monitoring, secure browser/provider abstraction where justified, incident separation and forensic logs.

Exit gate

- Exam survives network interruption without duplicate submission
- No actor can author, approve, mark and ratify contrary to policy
- Accommodation applies correctly without exposing diagnosis
- Moderation discrepancy workflow is complete and auditable
- Published result correction propagates to transcript safely
- Appeal preserves original decision and remediation history

Explicitly deferred unless required for the gate

- Unproven remote-proctoring claims
- Automatic final grading of high-stakes work
- Opaque plagiarism sanctions

Stage 6 - Ecosystem and credentials

Planning band: Months 18-24. Outcome: Make Veza portable, extensible and institution-friendly without weakening the canonical model.

Delivery scope

Work package	Detailed implementation
LTI 1.3/Advantage	Platform registration, OIDC launch, deep linking, names/roles, assignment/grade services, key rotation and tool governance.
OneRoster 1.2	CSV and REST adapter, mapping, delta sync, result ledger, conflict policy, retry and reconciliation.
Common Cartridge	Course/resource import-export, compatibility report, asset handling and round-trip regression suite.
Credentials	Credential templates, award rules, revocation, verification, Open Badges 3.0 and CLR 2.0 adapters.
Integration hub	Connector configuration, secrets, schedules, webhooks, mapping rules, status, retry, dead-letter and tenant support view.
Public API	Versioned REST API, OAuth clients, scopes, quotas, idempotency, OpenAPI documentation and sandbox.
Webhooks	Tenant subscriptions, signing, replay protection, delivery ledger, retries and schema versioning.
Developer portal	Documentation, test tenant, keys, changelog, status, sample applications and conformance evidence.

Exit gate

- Security review passes LTI launch and key rotation
- OneRoster sync is idempotent and reconciled
- Common Cartridge import reports loss instead of silent flattening
- Credential verification and revocation work independently
- API breaking-change policy is enforced
- Connector failures are visible and recoverable by authorised staff

Explicitly deferred unless required for the gate

- One-off connectors without reuse strategy
- Internal data-model changes solely to mirror partner APIs

Stage 7 - Intelligence and commercial scale

Planning band: Months 21-28. Outcome: Add governed intelligence, commercial controls and enterprise deployment options after the operational data is trustworthy.

Delivery scope

Work package	Detailed implementation
Learning data platform	Canonical event gateway, schema registry, streaming/archive, transformations, warehouse, semantic metrics and freshness monitoring.
Intervention analytics	Explainable signals, contributing factors, confidence, policy thresholds, case conversion and outcome feedback loop.
AI Assist	Tenant opt-in, provider abstraction, prompt/data policy, retrieval boundaries, human approval, citations, evaluation, cost limits and red-team testing.
Commercial metering	Entitlements, active-user and consumption measures, invoice ledger hooks, overage policy, credits and dispute evidence.
Marketplace	Tool/template catalogue, review, licensing, tenant approval, revenue share, abuse process and version compatibility.
Dedicated tiers	Automated silo/bridge provisioning, unified control plane, tenant rings, region selection, observability and upgrade orchestration.
Native mobile	Offline content contract, sync/conflict rules, push notifications, secure storage and parity decisions based on validated use cases.
Advanced reporting	Benchmarks, cohort comparisons, scheduled board reports and privacy-preserving aggregation.

Exit gate

- Metrics reconcile to source transactions
- Risk signals show factors and cannot auto-sanction learners
- AI output is labelled, evaluated and non-authoritative
- Usage ledger is reproducible for billing disputes
- Dedicated tenant is provisioned and upgraded automatically
- Offline mobile sync preserves attempts and completion

Explicitly deferred unless required for the gate

- AI decisions without human accountability
- Cross-tenant benchmarking without privacy thresholds
- Manual dedicated-tenant snowflakes

Stage 8 - GA hardening and rollout

Planning band: Months 26-30. Outcome: Prove that Veza can be sold, implemented, operated, supported and recovered repeatedly.

Delivery scope

Work package	Detailed implementation
Reliability programme	Load, soak, chaos, dependency failure, noisy-neighbour, backup/restore, multi-AZ and region recovery exercises.
Security assurance	Independent penetration test, dependency/container scanning, privileged-access review, incident tabletop and remediation closure.
Accessibility assurance	WCAG 2.2 AA audit across critical journeys, assistive technology testing, authoring and assessment evaluation.
Migration factory	Repeatable extract/map/validate/import/reconcile/cutover/rollback playbooks for priority source systems.
Implementation package	Discovery workbook, configuration catalogue, readiness checklist, training paths, admin certification and adoption plan.
Support model	Severity definitions, SLAs, escalation, status page, knowledge base, support access controls and tenant health monitoring.
Commercial readiness	Packaging, contracts, DPA, subprocessor register, pricing, sales engineering, demo tenants and procurement evidence.
Reference launches	At least two institutions complete migration, parallel run, go-live, hypercare and post-implementation review.

Exit gate

- SLOs sustained through representative load
- RPO/RTO recovery evidence signed
- Critical and high security findings closed
- WCAG audit blockers closed
- Two repeatable tenant go-lives completed
- Support, billing, status and escalation processes operational

Explicitly deferred unless required for the gate

- New major modules
- Nonessential visual redesign
- Unvalidated market expansions

4. Module implementation playbooks

The following playbooks define the minimum implementation envelope for major modules. Each module still requires detailed stories and prototypes, but these controls prevent “feature-complete” claims that omit institutional edge cases.

Module	Outcome	Core records	Non-negotiable invariants	Operational measures
Tenant and entitlement management	Provision, configure, meter, suspend, archive and delete contractual customers safely.	Tenant, plan, entitlement, quota, region, branding, domain, lifecycle state.	Idempotent provisioning; explicit lifecycle state machine; no hard delete while legal/academic retention applies; all product APIs consume resolved tenant context.	Provisioning success/failure, time to ready, entitlement denial, quota utilisation, tenant cost attribution.
Identity and membership	Give one person secure access to one or more institutions without merging tenant-controlled records.	User, credential, session, tenant membership, identity link, MFA method, recovery event.	Global identity does not imply tenant access; identity linking requires evidence; session carries active tenant; recovery cannot bypass MFA policy.	Sign-in success, recovery failure, suspicious session, invitation conversion, SSO fallback usage.
RBAC and policy engine	Authorise actions by permission, scope and contextual conditions.	Permission, role, assignment, scope, condition, policy decision, delegation.	Default deny; explicit tenant context; assignment expiry; delegated admin cannot grant beyond authority; sensitive policy decisions are auditable.	Denied actions, role changes, privilege escalation attempts, stale assignments, policy latency.
Institution and academic structure	Represent varied institution models without code forks.	Institution, campus, org unit, terminology, academic period, programme, offering, outcome.	Canonical names remain stable; labels are tenant-configurable; hierarchy and periods are effective-dated; deletion is constrained by dependent records.	Setup completion, hierarchy validation failures, terminology use, period conflicts.
People and relationships	Store institution-controlled person data and scoped relationships.	Person profile, learner/staff profile, identifier, relationship, consent, contact point.	Duplicate matching is reviewable; guardian/sponsor access is relationship-scoped and dated; sensitive fields use separate permissions.	Duplicate rate, merge reversals, relationship expiry, consent gaps.
Admissions	Move applicants through evidence-based decisions into identity and enrolment.	Application, form version, evidence, decision, offer, prerequisite check, applicant communication.	Form version preserved; decisions require authorised actor; rejected/waitlisted records retained by policy; accepted application conversion is idempotent.	Conversion rate, time in state, missing evidence, duplicate applicant, offer expiry.

Module	Outcome	Core records	Non-negotiable invariants	Operational measures
Catalogue, curriculum and course runs	Separate reusable academic design from each delivery instance.	Blueprint, version, module, activity, outcome mapping, approval, run, overlay, cohort.	Approved versions immutable; runs pin a version; upgrades use diffs and selective merge; overlays never mutate source.	Approval cycle, coverage gaps, upgrade adoption, orphaned outcomes.
Veza Studio	Author accessible, reusable and governable learning experiences.	Document tree, block, asset, template, comment, version, review, publish snapshot.	Autosave is not publish; published snapshot immutable; collaboration handles conflicts; unsupported imported content is reported; accessibility checks are visible and enforceable by policy.	Authoring errors, accessibility defects, publish failures, template reuse, review turnaround.
Learning delivery	Give learners a coherent, low-friction path through learning.	Course membership, module state, activity state, completion, bookmark, discussion, announcement.	Completion rules versioned; staff preview differs from learner state; hidden/future content not leaked; progress can be recomputed from authoritative events.	Activation, completion, stalled learners, content errors, bandwidth mode usage.
Calendar and timetable	Coordinate academic time, people and resources.	Calendar, event, recurrence, room/resource, participant, conflict, publication, change.	Time-zone semantics explicit; recurrence exceptions preserved; published changes notify affected users; clash overrides require reason.	Conflict rate, late changes, failed notifications, room utilisation.
Attendance	Create official attendance from reviewable evidence.	Session, attendance record, evidence, status, correction, approval.	Automated evidence is not official until policy allows/review occurs; corrections preserve original; QR/code resists replay; staff scope enforced.	Missing attendance, correction rate, suspicious scans, late submissions.
Assignments and submissions	Collect immutable evidence under transparent rules.	Assignment, attempt policy, submission attempt, file/text evidence, receipt, extension.	Each attempt immutable; server timestamp authoritative; reconnect is idempotent; extensions/accommodations do not expose sensitive source data.	Upload failures, duplicate attempts, late submissions, scan failures, receipt latency.
Assessment item bank and exams	Deliver reusable questions and reliable assessment sessions.	Item, item version, pool, test blueprint, form, exam session, response, incident.	Item exposure controlled; random forms remain equivalent; autosave/reconnect auditable; incidents separated from	Exposure, autosave lag, reconnect success, incident rate, form equivalence.

Module	Outcome	Core records	Non-negotiable invariants	Operational measures
			academic decisions.	
Marking, moderation and appeals	Produce defensible academic decisions.	Marker allocation, rubric score, feedback, moderation sample, discrepancy, ratification, appeal.	Separation of duties enforced; published results versioned; corrections and appeals never overwrite original; anonymous identities protected.	Marking turnaround, discrepancy, moderation coverage, appeal outcome, correction rate.
Gradebook and progression	Calculate and publish outcomes without silent changes.	Grade item, category, formula version, result, publication, progression decision.	Formula changes preview impact; rounding rules explicit; exemptions distinguish missing from excluded; publication is a state transition.	Calculation reconciliation, unpublished backlog, overridden results, progression exceptions.
Progress and interventions	Turn explainable signals into accountable support work.	Progress snapshot, signal, factor, case, action, owner, follow-up, outcome.	Signals show freshness/confidence; policy decides case creation; humans review context; sensitive notes are separated; outcomes improve evaluation.	Case response time, overdue actions, false-positive review, outcome completion.
Communications and notifications	Deliver timely, auditable messages through appropriate channels.	Template, message, audience, channel preference, delivery, read state, digest.	Academic/legal messages may override marketing preferences; recipient snapshot retained; retries deduplicated; guardian communications respect relationship scope.	Delivery failure, read rate, opt-out errors, duplicate sends.
Media and live learning	Deliver video and synchronous learning through provider abstraction.	Media asset, rendition, caption, transcript, live session, provider, recording.	Raw and processed assets isolated; signed URLs expire; captions editable; provider failure has fallback; recording consent and retention apply.	Transcode failure, playback startup, caption coverage, live join failures.
Credentials	Issue verifiable achievements based on approved results.	Credential template, award rule, issuance, evidence, verification, revocation.	Issuance references immutable result state; revocation does not erase history; verification exposes minimum necessary data; signing keys rotate safely.	Issuance failure, verification, revocation, wallet/export usage.
Integrations	Synchronise external systems without corrupting canonical state.	Connector, credential, mapping, sync job, source record, result ledger, conflict.	All writes enter domain commands; jobs idempotent; source identifiers preserved; conflicts require explicit	Sync lag, error rate, conflict backlog, retry success, stale credentials.

Module	Outcome	Core records	Non-negotiable invariants	Operational measures
			policy; dead letters visible.	
Analytics and reporting	Provide trustworthy operational and learning insight.	Event schema, metric, dimension, report, schedule, export, data quality issue.	Critical state comes from domain records/events; metric definitions versioned; freshness displayed; exports permissioned; cross-tenant aggregation thresholded.	Data freshness, reconciliation, report duration, failed schedules, metric adoption.
AI Assist	Assist users without becoming an unaccountable decision-maker.	AI policy, provider, prompt template, retrieval scope, output, feedback, evaluation.	Tenant opt-in; no training use without agreement; retrieval follows user permissions; citations/uncertainty visible; prohibited decisions blocked; cost and abuse limits enforced.	Groundedness, harmful output, human acceptance, override, latency, cost per task.

5. Technical implementation plan

5.1 Repository and application structure

Area	Recommended structure
Monorepo	pnpm workspaces with Turborepo or Nx; applications for web, API, workers, admin/control plane and documentation; packages for design system, domain contracts, authz client, observability, testing and configuration.
Frontend	Next.js App Router with server-rendered shells where useful, client islands for high-interaction Studio and marking, shared design-system package and typed API client generated from OpenAPI.
Backend	NestJS modular monolith organised by bounded context. No direct cross-context repository access; interactions use application services, published interfaces or events.
Workers	Dedicated worker processes for media, notifications, imports, exports, analytics, search indexing and connector synchronisation.
Schemas	PostgreSQL migrations are forward-compatible, reviewed and reversible through explicit remediation migrations. Event and API schemas are versioned in source control.
Feature flags	Flags have owner, purpose, tenant targeting, expiry date and removal task. Security controls cannot be bypassed by ordinary product flags.
Documentation	Architecture, ADRs, OpenAPI, event catalogue, runbooks, data dictionary and product behaviour documentation version with releases.

5.2 Bounded-context delivery order

Order	Context	Depends on	First usable slice
1	Tenant & Entitlements	None	Provision tenant with plan, region, modules and status.
2	Identity & Access	Tenant	Invite user, create membership, sign in, select tenant, authorise scoped action.
3	Audit & Compliance	Tenant + Identity	Record and query privileged action with retention class.
4	Institution Structure	Tenant + Access	Configure institution, units,

Order	Context	Depends on	First usable slice
			labels and academic period.
5	People & Relationships	Institution	Create learner/staff profile and scoped guardian relationship.
6	Academic Catalogue & Curriculum	Institution	Create programme/offering/outcomes and approved blueprint version.
7	Delivery & Enrolment	Catalogue + People	Create run/cohort and enrol learner.
8	Content & Studio	Curriculum + Access	Author, review and publish one accessible module.
9	Calendar/Timetable	Institution + Delivery	Publish class session and detect conflicts.
10	Assessment & Outcomes	Delivery + Content	Submit, mark, moderate and publish one result.
11	Progress/Intervention	Delivery + Assessment	Generate explainable signal and complete case.
12	Communications	People + Delivery	Send targeted announcement with delivery ledger.
13	Integrations	Core contexts stable	Execute one idempotent roster sync.
14	Analytics	Canonical events stable	Reconciled tenant operational dashboard.
15	Credentials	Approved results	Issue and verify one credential.

5.3 API and event standards

- Use REST/OpenAPI 3.1 for transactional public and internal APIs. GraphQL is limited to carefully selected aggregate read models where it materially improves client composition.
- Commands use idempotency keys where retries can create duplicate academic or financial state.
- Every response and log carries correlation, tenant and actor identifiers, with sensitive values redacted.
- Use optimistic concurrency or explicit version checks for collaborative and high-contention records.
- Publish domain events through a transactional outbox. Consumers must be idempotent and maintain replay checkpoints.
- Event schemas include tenant, aggregate, version, occurred-at, actor and trace metadata. Breaking event changes require a new version and migration window.
- Long-running imports, reports, media jobs and bulk operations are asynchronous with status, progress, cancellation and downloadable error detail.

5.4 Data architecture and isolation

Concern	Implementation requirement
Tenant key	All pooled tenant-owned records carry tenant_id and composite keys/indexes include it where appropriate.
Row-level policy	PostgreSQL row-level policies provide defence in depth; application authorisation remains mandatory.
Connection context	Database session context is set from trusted server-side tenant resolution and cleared between pooled connections.
Dedicated tier	Siloed data stores preserve the same logical schema, migrations, APIs and control-plane lifecycle.
Identifiers	External identifiers are namespaced by source and tenant. Public IDs are opaque; sequential IDs are not exposed.
Sensitive fields	Field-level encryption or separate stores for high-sensitivity records; keys and access are tenant/purpose aware.
Versioning	Effective-dated and immutable-version patterns apply to policy, curriculum, assessment, results and credentials.
Deletion	Deletion requests become policy-driven workflows with dependency analysis, legal holds, anonymisation and verified destruction.
Reporting	Operational read models and warehouse tables are derived; transactional tables remain authoritative.
Test data	Synthetic tenant-safe data only outside authorised migration environments; production copies are prohibited by default.

5.5 Infrastructure environments

Environment	Purpose	Data and access rules
Local	Developer feedback and unit/component tests.	Synthetic seeds only; no cloud credentials required for ordinary work.
Ephemeral preview	Per-PR integration and visual review.	Short-lived, synthetic, isolated; automatic teardown and cost caps.
Integration	Cross-team contracts, migrations and asynchronous workflow tests.	Synthetic scale datasets; stable external sandboxes.
Staging	Production-like release candidate, performance and migration rehearsal.	Masked or synthetic data; production topology and security controls.
Pilot production	Design-partner operations with tenant-ring	Real tenant data, contractual controls,

Environment	Purpose	Data and access rules
	controls.	restricted support access and enhanced monitoring.
Production shared	General SaaS pooled/bridge tiers.	Strict tenant isolation, SLOs, backups, WAF, incident response and cost allocation.
Production dedicated	Contracted protected/sovereign tier.	Automated isolated resources managed through the same control plane and release process.
DR	Recovery capability and exercises.	Encrypted replicated backups/configuration; access only for rehearsals or incidents.

5.6 Observability and service levels

Signal	Required implementation
Logs	Structured, tenant-aware, correlation-linked, privacy-filtered and retention-controlled.
Metrics	Golden signals plus domain measures such as submission receipts, grade publication, sync lag and media readiness.
Traces	End-to-end request and asynchronous trace propagation through API, queue, worker and external connector.
Audit	Separate immutable business/security audit from operational logs.
SLOs	Define availability and latency objectives per critical journey, not only per component.
Alerting	Actionable symptom-based alerts tied to runbooks, owners and tenant impact.
Synthetic monitoring	Continuous sign-in, course access, submission, marking and integration probes using synthetic tenants.
Cost telemetry	Tenant/tier attribution for compute, storage, media, search, AI and integration jobs.
Status communication	Public status page and tenant-specific incident notices with clear impact and recovery updates.

6. Product design and UX implementation plan

6.1 Information architecture

The application shell should be role-adaptive but structurally consistent. Global navigation handles tenant switch, search, notifications, help and profile. The left workspace navigation holds stable product areas. The centre canvas contains the active task. A right contextual panel appears only for properties, collaborators, learner context, review or history. Full-screen modes are reserved for Studio, live teaching, exam delivery and complex marking.

Experience	Primary navigation emphasis	Default landing outcome
Learner	Today, Courses, Calendar, Messages, Progress and Credentials.	What requires attention now, what is next and whether anything blocks progress.
Instructor	Today, Classes, Content, Assessments, Marking, Learners and Calendar.	Sessions, preparation, unmarked work, attendance gaps and support needs.
Academic operations	People, Admissions, Academic Setup, Timetable, Attendance, Results, Reports and Cases.	Exceptions, deadlines, capacity, missing data and approvals.
Course designer	Studio, Templates, Assets, Reviews, Outcomes and Published Courses.	Work in progress, reviews, accessibility issues and reusable content.
Tenant administrator	Setup, Identity, Roles, Integrations, Branding, Policies, Usage, Audit and Support.	Readiness, configuration health, security, usage and unresolved operational issues.
Veza operator	Tenants, Plans, Provisioning, Regions, Incidents, Releases, Usage and Support.	Fleet health, tenant risk, release rings and operational exceptions.

6.2 Bento-minimal design rules

- Bento surfaces group related operational signals, not decorative cards. Each tile must answer a decision or lead to a task.
- Use one dominant work area and a limited number of secondary modules; avoid equal-weight dashboards that force users to scan everything.
- Colour communicates state and hierarchy. Brand colour is not used as an indiscriminate background fill.
- Dense administrative tables support column control, saved views, filters, bulk selection, keyboard navigation and export permissions.
- Learner experiences favour calm progressive disclosure, clear next action, visible effort and low cognitive load.
- High-stakes modes remove nonessential navigation, show autosave/connectivity, clarify time and preserve recovery routes.
- All destructive or consequential actions show scope, impact, reversibility and audit consequence before confirmation.

6.3 Visual system direction

Token family	Direction
Core ink	Deep blue-charcoal for text and structural navigation; avoids harsh pure black while preserving authority.
Brand teal	Reserved for primary actions, selected states and progress emphasis; chosen to feel modern, trustworthy and distinct from warning/success states.
Canvas	Warm off-white/very light neutral to reduce glare across long learning sessions.
Surface	White and subtle neutral elevations; borders and spacing carry structure more than shadows.
Success	Muted green with icon/text; never colour alone.
Warning	Amber for attention and pending institutional decisions.
Critical	Controlled red for errors, risk and destructive actions.
Information	Blue for neutral guidance and system notices.
Typography	Highly legible sans-serif with clear numeric forms; display weight used sparingly for hierarchy.
Density	Comfortable default, compact admin mode and distraction-reduced learner/exam mode.

6.4 Research and validation cadence

Moment	Research method	Minimum evidence
Discovery	Contextual interviews, workflow observation, artefact review.	At least 5 participants per priority role across design partners before locking major workflows.
Concept	Clickable prototypes and task-based usability.	Critical task success $\geq 80\%$ without facilitator intervention before build commitment.
Build	Weekly design QA and accessibility review against implemented slices.	No unresolved severity-1 UX/accessibility defect in pilot scope.
Pilot	Telemetry, interviews, support-ticket analysis and shadowing.	Measured time-on-task, error rate, adoption and qualitative confidence.
Release	Accessibility regression, content review and workflow sign-off.	Evidence stored with release readiness pack.
Post-launch	30/60/90-day adoption and outcome review.	Prioritised remediation and validated next-phase assumptions.

7. Quality engineering and validation plan

7.1 Test layers

Layer	Purpose	Release expectation
Static quality	Type checking, linting, dependency policy, secret scanning and schema validation.	Required on every change.
Domain unit tests	Prove invariants such as enrolment, grading, progression, permissions and retention.	High coverage of decision logic; branch coverage targets are risk-based.
Property-based tests	Explore large input spaces for formulas, schedules, randomised exams and tenant isolation.	Mandatory for critical mathematical and combinatorial logic.
Component tests	Accessible UI behaviour, forms, editor blocks and error states.	All design-system components and critical domain widgets.
Contract tests	API, event and connector compatibility.	Producer and consumer tests block incompatible release.
Workflow tests	Complete journeys across contexts.	Admissions, enrolment, teaching, submission, marking, moderation, results, appeal and credentials.
Permission tests	Permission x role x scope x condition matrix.	Automated generated tests including deny, expiry and delegated authority.
Migration tests	Schema, tenant data, imports and rollback/reconciliation.	Every migration tested against representative scale and prior version.
Performance tests	Load, stress, soak, concurrency and large-job fairness.	Critical journey SLO and noisy-neighbour evidence before pilot/GA.
Resilience tests	Dependency failure, queue backlog, provider outage, reconnect, backup and DR.	Scheduled game days and release-specific failure tests.
Security tests	SAST, DAST, dependency/container, IaC, auth abuse and penetration testing.	No open critical; high findings require explicit board-approved treatment.
Accessibility tests	Automated, keyboard, screen reader, zoom/reflow, captions and exam usability.	WCAG 2.2 AA target across critical journeys.
Visual regression	Breakpoints, density modes, themes and long/localised content.	Design-system and critical screens on each release train.

7.2 Golden workflow suites

Suite	Required scenarios
Tenant lifecycle	Provision, failed provision/retry, plan change, module removal, suspension, export, archive and deletion hold.

Suite	Required scenarios
Identity and access	Invite, SSO link, multi-tenant switch, MFA recovery, role expiry, delegated admin denial and assisted session.
Academic setup	Create hierarchy/period/catalogue, approve blueprint, create run, enrol, transfer, withdraw and close run.
Authoring	Create template, collaborate, accessibility issue, review, approve, publish, update blueprint and merge run overlay.
Daily teaching	Launch class, attendance, material release, announcement, learner concern and follow-up case.
Assessment	Create assignment, accommodate, submit/reconnect, mark, moderate, release, correct and appeal.
Exam	System check, timed attempt, disconnect, incident, time adjustment, submit receipt, ratify and release.
Integration	Initial import, delta sync, duplicate delivery, conflict, retry, dead letter, credential rotation and reconciliation.
Credential	Eligibility, issue, verify, export, revoke and reissue.
Offboarding	Course archive, learner export, tenant data retention, identity unlink and verified destruction.

7.3 Defect severity

Severity	Definition	Release rule
S0 - Security/data integrity	Cross-tenant exposure, unauthorised privileged action, irreversible result corruption, credential forgery or data loss.	Immediate stop-ship and incident process.
S1 - Critical workflow	Critical learner/institution journey cannot complete or recovery is unsafe.	Stop-ship for affected scope.
S2 - Major	Material workaround, incorrect result/notification, serious accessibility failure or operational instability.	Must be fixed or formally removed from release scope.
S3 - Moderate	Usability or functional defect with safe workaround.	May ship with owner and dated remediation.
S4 - Minor	Cosmetic or low-impact inconsistency.	Prioritised through normal backlog.

8. Security, privacy and compliance implementation

8.1 Security work programme

Control area	Implementation activities
Secure SDLC	Threat modelling at epic inception, secure coding standards, code owners, dependency policy, automated scanning, security test cases and remediation SLAs.
Tenant isolation	Trusted tenant resolution, policy enforcement, RLS defence, storage namespaces, cache/search keys, queue metadata, test generators and dedicated-tier automation.
Authentication	MFA policies, session rotation, device/session view, risk signals, SSO hardening, recovery controls and break-glass governance.
Authorisation	Central policy decision library/service, deny-by-default, scope validation, field permissions, delegated administration and exhaustive matrix tests.
Privileged access	Just-in-time access, approval where required, assisted-session restrictions, step-up authentication, immutable audit and periodic review.
Data protection	Classification, encryption in transit/at rest, field encryption, KMS separation, secrets management, backups, retention and destruction evidence.
Application security	Input validation, output encoding, CSRF/XSS/SSRF controls, rate limits, upload quarantine, malware scanning and secure headers.
Assessment security	Item access controls, exposure telemetry, role separation, secure delivery modes, incident evidence and controlled exports.
Vendor risk	Due diligence, DPAs, subprocessor register, data-flow review, breach terms, key rotation, exit plan and annual reassessment.
Incident response	Tenant-aware triage, evidence preservation, containment, notification workflow, status communication, lessons learned and control updates.

8.2 Privacy-by-design delivery checklist

- Every data field has purpose, owner, sensitivity, legal basis/authority, retention class and visibility rule.
- Optional data collection is disabled until a tenant enables a justified use case.
- Children, guardian authority and age-appropriate notices are modelled explicitly where relevant.

- Data-subject access, correction, restriction, export and deletion are operational workflows with academic/legal exceptions.
- Cross-border processing and regional placement are tenant-configurable only where contracts and providers support them.
- Analytics, AI and support tooling receive the minimum fields required and preserve tenant/user permissions.
- Production support access is time-limited, purpose-bound and visible in audit logs.
- Tenant offboarding includes export, retention hold, revocation, deletion schedule and destruction certificate.

8.3 Compliance evidence pack

Evidence	Owner	Refresh
Data inventory and flow map	Privacy + architecture	Each major release
DPIA and risk treatment	Privacy/security	At least annually and on material change
Access-control matrix and test report	Security + QA	Each release
Penetration test and remediation	Security	Before GA and annually
Backup/restore and DR evidence	SRE	Quarterly restore; annual DR
Accessibility conformance report	Design/QE	Major releases
Subprocessor register and vendor reviews	Legal/privacy	Continuous; annual review
Incident tabletop and lessons	Security/operations	Twice yearly
Retention/destruction verification	Data governance	Quarterly sample
Tenant isolation test evidence	Engineering/QE	Each release

9. Institution implementation, migration and rollout

9.1 Standard institution onboarding lifecycle

Stage	Activities	Exit criterion
1. Qualification	Segment, use cases, users, modules, integrations, data volumes, residency, SSO, academic calendar and decision authority.	Scope and commercial fit confirmed.
2. Discovery	Current-state workflows, source systems, data quality, terminology, policy variance, reports, accessibility and change impacts.	Signed solution blueprint and migration inventory.
3. Configuration	Tenant, branding, terminology, academic structure, policies, roles, templates, identity and integrations.	Configuration validation passes.
4. Migration rehearsal	Extract, map, cleanse, dry-run import, reconcile, exception resolution and rollback.	Reconciliation thresholds met and cutover plan approved.
5. Training	Role-based administrator, instructor, support and learner readiness; sandbox practice and certification.	Named operational owners pass readiness checks.
6. Pilot/parallel run	Limited cohort or programme, monitored workflows, support war room and daily reconciliation.	Pilot KPIs and defects meet go-live threshold.
7. Cutover	Change freeze, final delta, validation, communications, switch-over and rollback window.	Business owner signs production acceptance.
8. Hypercare	Enhanced monitoring, daily triage, adoption support and issue correction.	Stable operations for agreed period.
9. Transition	Handover to customer success/support, optimisation backlog and success plan.	30/60/90-day outcomes scheduled.

9.2 Migration workbench requirements

- Source connector or secure file intake with checksum and chain of custody.
- Canonical mapping templates for people, identifiers, relationships, academic structures, offerings, runs, enrolments, content, grades and attendance.
- Dry-run validation with counts, referential integrity, duplicate candidates, invalid values and policy conflicts.
- Idempotent import batches with source IDs, status, error rows and retry.
- Reconciliation dashboards comparing source totals, target totals and sampled record detail.
- Content compatibility report for imported packages, links, media and assessment formats.
- Cutover delta strategy and explicit freeze window.
- Rollback or compensating plan for each imported domain.
- Signed migration report retained in tenant audit records.

9.3 Readiness scorecard

Domain	Go-live checks
Identity	Admin and learner sign-in, SSO/MFA, recovery, break-glass and deprovisioning tested.
Academic setup	Periods, programmes, courses, cohorts, staff allocations and enrolments reconciled.
Teaching	Course content, sessions, resources, announcements and attendance workflow tested.
Assessment	Assignments/exams, accommodations, rubrics, grade formula, moderation and release policy approved.
Communications	Templates, sender domains, delivery channels, escalation and guardian rules tested.
Data/privacy	Notices, consent, retention, exports, sensitive access and support permissions approved.
Integrations	Initial and delta sync, credential rotation, failure alert and reconciliation owner established.
Operations	Support contacts, severity model, status page, diagnostics, runbooks and on-call readiness.
Training	Tenant admins, instructors and helpdesk completed required learning paths.
Continuity	Backup, export, outage procedures, provider fallback and rollback window agreed.

10. Commercial and customer-operations implementation

10.1 Packaging model

Layer	Implementation
Platform edition	Core tenancy, identity, academic structure, Studio, learning delivery, assignments, gradebook, calendar, communications and baseline reporting.
Operational modules	Admissions, timetable, attendance, guardian/sponsor, intervention/case management, advanced reports and credentials.
Assessment module	Item bank, exams, moderation, ratification, appeals and assessment security.
Integration module	SSO/SCIM, OneRoster, LTI, APIs, webhooks and premium connectors.
Intelligence module	Advanced analytics, intervention models and governed AI Assist.
Deployment tier	Shared, protected bridge/silo and dedicated/sovereign options under one control plane.
Consumption	Storage, media processing/delivery, messages, AI and exceptional integration jobs measured through auditable ledgers.
Services	Implementation, migration, configuration, training, custom connector development and premium support separately scoped.

10.2 Customer health model

Dimension	Signals
Technical health	Error rate, integration lag, identity failures, quota pressure, support incidents and configuration drift.
Adoption	Active learners/instructors, course activation, content publishing, assessment use and module adoption.
Operational maturity	Unresolved setup checks, role sprawl, overdue cases, manual imports and policy exceptions.
Learning operations	Submission completion, marking turnaround, attendance completeness and intervention follow-through.
Commercial	Entitlement utilisation, consumption trend, renewal date,

Dimension	Signals
	service usage and invoice disputes.
Relationship	Executive sponsor engagement, admin champion health, open escalations and roadmap alignment.

11. Work breakdown structure and backlog hierarchy

11.1 Backlog levels

Level	Definition	Example
Outcome	Business or institutional capability proven in production.	Institution can conduct a moderated assignment and publish defensible results.
Programme increment	Multi-team release objective lasting 8-12 weeks.	Assessment foundation and gradebook versioning.
Epic	Coherent capability with one accountable product owner.	Rubric authoring and scoring.
Feature slice	End-to-end user-visible or operational slice.	Instructor creates criterion and learner views rubric.
Story	Small independently testable behaviour.	Prevent deletion of criterion used by published rubric.
Task	Implementation activity that does not independently deliver behaviour.	Add database constraint and migration.
Enabler	Architecture, security, research or platform work required by future slices.	Policy decision cache with tenant-safe keys.

11.2 First twelve programme increments

Increment	Primary objectives	Demonstration
PI-01	Mobilisation, research, domain map, design shell, repo and CI.	Synthetic user signs into shell; ADR and test systems operate.
PI-02	Tenant provisioning, memberships, base permissions, audit and environments.	Create tenant, invite admin, deny cross-tenant access and inspect audit.
PI-03	Institution structure, academic periods, people and terminology.	Configure two different institution models without code change.
PI-04	Catalogue, outcomes, blueprints, runs, cohorts and enrolment.	Create approved course run and enrol learner with conflict validation.
PI-05	Studio editor, assets, review and publication.	Author accessible module, approve and publish immutable version.
PI-06	Learner course room, calendar, announcements and progress.	Learner completes module on mobile and sees next action.
PI-07	Assignments, submission receipts, marking and basic gradebook.	End-to-end submission to released result with audit.
PI-08	Teaching workspace, attendance and communications.	Instructor conducts class and resolves attendance exception.

Increment	Primary objectives	Demonstration
PI-09	Admissions, bulk imports, timetable and SSO.	Applicant converts to enrolled SSO learner through rehearsed import.
PI-10	Interventions, guardian/sponsor and reports.	Risk signal becomes owned case and scoped guardian summary.
PI-11	Item bank, exams, moderation and appeals.	Controlled exam survives reconnect and result passes ratification.
PI-12	Standards adapters, credentials and GA migration factory.	OneRoster/LTI integration plus verifiable credential and tenant cutover.

11.3 Definition of ready

- User and institutional problem is evidenced and priority is approved.
- Primary and edge-case flows are mapped, including failure and recovery.
- Affected roles, scopes and policy conditions are identified.
- Domain ownership and data changes are clear; unresolved ADRs are listed.
- Privacy, security, accessibility, analytics and operational requirements are specified.
- External dependency, migration and rollout assumptions are known.
- Acceptance criteria are testable and the slice can be delivered vertically.

11.4 Definition of done

- Behaviour works through supported UI and API paths and meets acceptance criteria.
- Domain invariants, permission checks, audit and tenant isolation are covered by automated tests.
- Accessibility, responsiveness, empty/loading/error and low-bandwidth states are complete.
- Events, metrics, logs, traces and support diagnostics are implemented.
- Migrations are tested; rollback or remediation procedure exists.
- Threat-model and privacy deltas are reviewed and required controls closed.
- Documentation, help content and runbooks are updated.
- Feature flag, rollout ring, monitoring and removal/cleanup plan are defined.
- Product, design, QA and engineering owners accept the evidence.

12. Programme risks and mitigations

ID	Risk	Rating	Mitigation
R1	Scope expansion across institution types	High	Prioritise canonical workflows and terminology configuration; use design partners; prohibit bespoke forks.
R2	Building generic LMS screens before academic model	Critical	Make academic graph and domain contracts the critical path; architecture council blocks parallel shadow models.
R3	Tenant data leakage	Critical	Trusted context, RLS defence, scoped storage/cache/search, generated isolation tests, pen testing and stop-ship policy.
R4	RBAC role explosion	High	Permission + scope + policy model, baseline roles, tenant custom roles with constraints and periodic access review.
R5	Assessment disputes and result corruption	Critical	Versioned formulas/results, separation of duties, immutable evidence, moderation and controlled corrections/appeals.
R6	Migration data quality	High	Discovery inventory, dry runs, source IDs, reconciliation, exception ownership and rollback.
R7	Integrations dominate roadmap	High	Canonical adapter framework, prioritised standards, connector commercial model and strict custom-integration intake.
R8	Low adoption by instructors	High	Co-design, Today workspace, template-first Studio, training, in-product guidance and measured task time.
R9	Learner connectivity constraints	High	Responsive web, resumable uploads, adaptive media, lightweight modes, autosave and offline-oriented contracts.
R10	Over-engineering microservices	Medium	Modular monolith first; extract only against measured scale, isolation or ownership needs.
R11	AI trust and privacy failure	High	Opt-in, governed provider abstraction, permissions-aware retrieval, evaluation, human approval and prohibited decision policy.

ID	Risk	Rating	Mitigation
R12	Dedicated tenants become snowflakes	High	Unified control plane, IaC, common release train, automated provisioning and explicit exception sunset.
R13	Accessibility debt	High	Accessible design system, authoring checks, automated/manual test gates and specialist review from Stage 0.
R14	Operational support overload	High	Diagnostics, tenant health, runbooks, staged rings, pilot caps and support staffing before expansion.
R15	Commercial metrics cannot be defended	Medium	Immutable usage ledger, metric definitions, reconciliation and customer-visible receipts.

12.1 Trigger-based contingency decisions

Trigger	Required decision
Pilot workflow failure rate exceeds threshold	Freeze expansion, root-cause by workflow, reduce scope or extend pilot until evidence recovers.
Cross-tenant defect discovered	Invoke security incident process, suspend affected feature, assess all tenants and require isolation regression before release.
Migration reconciliation below agreed threshold	Do not cut over; correct source mapping/data or narrow migrated scope.
Reference tenant requests bespoke code	Classify as configuration, reusable extension, connector or out-of-scope; commercial and architecture review required.
SLO missed under representative load	Profile and remediate before onboarding more tenants; introduce quotas/fairness or topology changes as needed.
AI evaluation below quality/safety threshold	Keep capability internal or disabled; do not compensate with disclaimers alone.
Critical vendor dependency fails assurance	Select alternate provider or isolate behind abstraction with exit path before production use.

13. Programme measurement framework

13.1 Product and institutional outcomes

Outcome	Candidate measures
Faster institution setup	Days from contract to configured tenant; setup-check completion; migration exception rate.
Efficient teaching operations	Instructor time to prepare class; attendance completion; marking turnaround; communication delivery.
Learner clarity and persistence	Next-action comprehension, activation, on-time submission, course progress and support response.
Defensible assessment	Moderation coverage, discrepancy, correction, appeal cycle time and audit completeness.
Intervention effectiveness	Signal review time, case ownership, follow-up completion and recorded outcomes.
Platform trust	Availability, submission receipt success, data reconciliation, security incidents and support satisfaction.
Portability	Import success, compatibility loss, connector reliability, API adoption and credential verification.
Commercial health	Time to value, active usage, module expansion, retention, cost-to-serve and gross margin by tier.

13.2 Delivery health

Measure	Interpretation
Gate evidence complete	Primary readiness measure; milestones are not accepted on narrative status alone.
Lead time for change	Shows flow from approved slice to production.
Change failure rate	Measures deployment quality and release safety.
Escaped S0-S2 defects	Indicates risk control effectiveness.
Rework from ambiguous requirements	Signals discovery and acceptance-quality problems.
Cross-team blocked time	Reveals unstable contracts or sequencing errors.
Architecture exception age	Prevents temporary shortcuts becoming permanent.
Test suite duration/flakiness	Protects delivery feedback speed.

Measure	Interpretation
Cloud cost per active tenant/user	Supports commercial and technical efficiency.
Support cases per active user	Tracks usability, reliability and implementation quality.

14. General availability launch gate

Category	Mandatory evidence
Product	Committed GA workflows complete; known limitations documented; no customer-specific fork required for reference tenants.
Architecture	Critical ADRs current; bounded-context ownership enforced; migrations and rollback reviewed.
Security	Independent test complete; no critical/high untreated risk; incident and privileged-access controls rehearsed.
Privacy/legal	DPA, privacy notices, data flows, subprocessors, retention, subject-rights and cross-border controls operational.
Accessibility	Critical journeys meet WCAG 2.2 AA target with manual assistive-technology evidence.
Quality	Golden workflows pass; performance, soak, resilience, permission and tenant-isolation suites pass.
Reliability	SLOs sustained; backups/restores verified; RPO/RTO and dependency fallback rehearsed.
Data	Migration factory reconciles reference tenants; warehouse metrics reconcile to source; deletion/offboarding tested.
Operations	24/7 incident ownership where contracted, runbooks, dashboards, status page, support access and escalation ready.
Customer readiness	Training, knowledge base, admin certification, adoption metrics and customer-success plans complete.
Commercial	Plans, entitlements, usage ledger, invoicing hooks, contracts and sales engineering assets complete.
Release	Canary/rings, feature flags, rollback, communication and post-release monitoring approved.
Executive approval	Product, architecture, security, operations and commercial owners sign the launch record.

14.1 First 90 days after GA

Window	Focus
Days 0-14	Hypercare, daily operational review, migration

Window	Focus
	reconciliation, tenant feedback and defect triage.
Days 15-30	Stability remediation, adoption interventions, documentation gaps and support trend analysis.
Days 31-60	SLO and cost review, entitlement/usage validation, customer health baselines and roadmap correction.
Days 61-90	Formal post-launch review, architecture debt plan, pricing/packaging evidence, reference case studies and next expansion gate.

Appendix A - RACI for critical outcomes

Outcome	A	R	C	I
Product scope and phase gate	Product director	Domain PMs	Architecture, design, security, delivery	All teams
Domain architecture	Chief architect	Technical leads	Product, data, security	Engineering
UX and accessibility	Design director	Product designers/QE	Academic group, engineering	Product teams
Security acceptance	Security lead	Engineering/SRE/QE	Privacy/legal, architecture	Executive sponsor
Migration acceptance	Implementation lead	Implementation/data engineers	Tenant owner, product, support	Customer success
Release go/no-go	Release readiness board	Release manager	All workstream leads	Tenants
Academic policy validity	Academic product specialist	Domain PM	Advisory group, tenant academic owner	Engineering
Operational SLOs	SRE lead	Platform/SRE	Engineering, support, product	Executive sponsor

Appendix B - Initial architecture decision backlog

ADR	Decision
ADR-001	Canonical tenant context propagation and trust boundary.
ADR-002	Pooled, bridge and silo data-topology rules.
ADR-003	Global identity versus tenant-owned person profiles.
ADR-004	Permission, scope and policy-condition representation.
ADR-005	Modular monolith boundaries and cross-context communication.
ADR-006	Transactional outbox and event schema/versioning.
ADR-007	Course blueprint/run overlay and upgrade merge model.
ADR-008	Assessment attempt immutability and result versioning.
ADR-009	File upload quarantine, scanning and media pipeline.
ADR-010	Calendar recurrence and timezone semantics.
ADR-011	Search indexing with permission-aware filtering.
ADR-012	Analytics event gateway and warehouse ownership.
ADR-013	Integration connector framework and secrets lifecycle.
ADR-014	Offline/resumable learner interaction contracts.
ADR-015	AI provider abstraction, governance and evaluation.
ADR-016	Dedicated tenant provisioning under unified control plane.
ADR-017	Data retention, deletion, archive and legal-hold model.
ADR-018	Credential signing, verification and key rotation.
ADR-019	Public API versioning and deprecation policy.
ADR-020	Feature flags, tenant rings and release compatibility.

Appendix C - Source architecture traceability

This implementation plan preserves and operationalises the source architecture's product thesis, tenant model, academic domain, scoped RBAC, user journeys, modules, technical stack, bounded contexts, standards strategy, privacy controls, non-functional requirements, test strategy, failure scenarios and phased roadmap. The source architecture remains authoritative for target product behaviour where this plan does not explicitly refine execution order or delivery gates.

Appendix D - Standards and implementation references

Reference	Implementation relevance
AWS SaaS Lens - Tenant Isolation	Tenant isolation is foundational and must be enforced across application and infrastructure layers.
AWS SaaS Architecture Fundamentals - Control Plane vs Application Plane	Supports separating global tenant lifecycle/operations from tenant-facing application capabilities.
1EdTech Interoperability Standards	Source catalogue for LTI, OneRoster, QTI, Common Cartridge, Caliper, Open Badges and related conformance work.
1EdTech LTI 1.3 / LTI Advantage	Secure external-tool launch, deep linking, roster/role and assignment/grade services.
1EdTech OneRoster 1.2	Roster, class, course and grade exchange.
1EdTech QTI 3	Portable assessments with accessibility and internationalisation support.
W3C WCAG 2.2	Accessibility target for web content and critical LMS workflows.
NestJS official documentation	Reference for modular backend, OpenAPI, testing and asynchronous/microservice patterns where extraction is justified.
OWASP ASVS and API Security guidance	Application and API security verification baseline.
South African POPIA and Information Regulator guidance	Privacy, operator, breach, data-subject and cross-border implementation requirements for South African deployments.